

The Summary of Underwater Gliders Control Strategies

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Abstract. The underwater glider is a new type of system that combines buoy technology and underwater robotics technology. By changing the net buoyancy as a driving force and changing its own center of gravity position, the attitude angle can be changed. Many advantages, such as gliding with high efficiency, wide cruise range, less power consumption, low noise, and no pollution, make the underwater glider an important platform for marine environment observation and ocean resource exploration. In this paper, control strategies for existing underwater gliders are reviewed. A total of 51 papers indexed by Scopus with keywords control and underwater gliders were reviewed from 1989 to 2016. The majority of gliders use classical controllers, which cannot dynamically compensate for un-modeled hydrodynamic forces and unknown variations in water current and wind. With increasing operational depths and larger payloads, control strategies will become an increasingly important aspect for these gliders. Control strategies implemented in underwater gliders were reviewed and alternative control strategies are proposed.

Keywords: Underwater gliders · Control strategies · Summary

1 Introduction

An underwater glider is a type of autonomous underwater vehicle (AUV) that uses buoyancy control in conjunction with wings to convert vertical motion to horizontal motion in a saw-tooth pattern and thereby propelling itself in forward direction with very low power consumption [1]. They use buoyancy and wings to produce forward motion for long duration ranging from weeks to months and cover distances over 3000 km [2]. There are more than 50 different AUGs developed for research and commercial purposes. Much theoretical and experimental analysis has been carried out on glider dynamics [3, 4], glider design and development for payload and maneuverability [5, 6], performance and stability analysis [7, 8], trajectory control analysis [9, 10] and path planning [9, 11] based on different controller techniques [12]. The majority of these gliders use classical controllers such as Proportional Integral-Derivative (PID) and Linear Quadratic Regulator (LQR). However, These control methods cannot provide

simultaneously easy control implementation and fast convergence speed for stabilization in the presence of continuously varying water current [13].

In this paper, existing underwater glider control strategies are reviewed with emphasis on their actuation system that includes wings and rudder control, followed by external effects of water current and variable loads which influence their motion.

No work to date achieved the control of fully under actuated vehicles via dynamic modelling with the presence of external varying forces such as water current and advanced controller design to compensate these problems and then implement them to underwater gliders. In order to overcome these issues, serious design and implementation considerations are required for robust control approach.

2 Underwater Glider Motion Control

The motion of an underwater gliders depends upon a buoyancy control system combined with a set of wings and rudder to move in vertical and horizontal direction through a water column [14]. The glider descends and ascends in water column due to control of buoyancy engine, which consists of oil filled bladder or piston evacuated cylinder. Buoyancy is controlled to be slightly positive or negative due to which a glider can move with almost zero power consumption with limited speed [15]. For example, the Spray glider as shown in Fig. 1, has a maximum speed of 0.35 m/s but can operate for up to six months.



Fig. 1. 1-Slocum [8], 2-Spray [16], 3-Seaglider [8]

Regardless of the design, gliders depend on vertical space to operate efficiently but cannot resist with strong currents, which ultimately creates a problem to control motion and trajectory. On the other hand, the average horizontal speed of underwater gliders is about 0.3 m/s. The low speed limitation has created challenges in controller performance such as control of vehicle orientation, position and velocity [14]. There is a direct effect of environmental forces including winds, wave and surface currents upon the motion of underwater gliders and AUVs [21]. These effects may be compensated through an effective control strategy. Therefore the control strategy should be chosen to reduce the effect of these disturbances. However, the challenges associated with control of underwater gliders include a highly nonlinear dynamic system and complex hydrodynamic effects. Different control strategies for laboratory scale and commercial scale have been implemented to underwater vehicles which are discussed in Table 1.

Table 1. Laboratory scale gliders control controllers

Type of Glider	Institution	Micro controllers	Speed & depth	Remarks
ALBAC glider	University of Tokyo	Two NEC V50 CPUs controlled by math processor Intel 80 [8]	0.5 to 1.0 m/s, 300 m	Trajectory is controlled by changing pitch and roll angle
ZJU glider	Zhejiang University (ZJU) glider	TMS320F28335 controller for real time motion control [4]. AMESim software is used to investigate dynamic performance of buoyancy control system	0.6 m/s, 200 m	Dynamic analysis, stability analysis and root locus for varying ballast mass for pitch control are performed by ZJU glider
Sea-Wing underwater glider	Shenyang Institute of Automation	NXP LPC series ARM processor chip with ARM7TDMI-S based high-performance 32-bit RISC Microcontroller [17]	0.8 m/s, 1200 m	The buoyancy-regulating mechanism alters the buoyancy of underwater glider by feeding hydraulic oil into or bleeding hydraulic oil out of the external oil bladder to produce the driving buoyancy
Seaglider	University of Washington	Onset Computer Corporation's TT8 controller. Motorola MC68332 microcontroller combined with a 12-b A/D converter [18]	0.25 m/s, 1500 m	Pitch and roll are controlled by using 16 mm Maxon neodymium magnet motors
Alex glider	Department of Marine System engineering Osaka Prefecture University	MAVC2 (Micro Aerial Vehicle Controller which has the ability to control 12 servo motors and a program to control the motion of glider is stored in this controller [7])	0.6 m/s, 200 m.	Pitch and roll motion is controlled by twisting of wings and steering of rudder
USM glider	USM	Arduino Mega (Primary controller) PC 104 (Secondary controller) LQR, MPC, and NNPC. [19]	0.4 m/s, 30 m	Internal sliding mass and ballast pump which are used as internal actuators control the pitch and buoyancy. Wings and rudder control maneuverability

(continued)

Table 1. (continued)

Type of Glider	Institution	Micro controllers	Speed & depth	Remarks
Miniature underwater glider prototype (A robotic Fish (Grace))	Developed in the Smart Microsystems Laboratory at MSU	dsPIC30F6014A provides storage memory. IMU including gyros(ST LPY503AL), accelerometer and digital compass (ST LSM303DLMRT) are also equipped [13, 20]	0.36 m/s, 20 m	Movable battery mechanism for Pitch control. Pump is used with internal cylindrical tank system. For sealing O-rings and for wings Aluminum wings are used
FOLAGA	Combined research effort: IMEDEA, ISME and University of Genova, Italy [8]	PC104 card, GPS receiver, a depth sensor inclinometer (pitch)	1.01 m/s, 50 m	A robust PD control system followed by back-stepping methodology is implemented by FOLAGA
OUC glider	Ocean University of China	STM32F107 controller for underwater glider based high-performance 32-bit RISC Microcontroller [51]	1.5 m/s, 1500 m	Fuzzy-PID controller was designed to control the pitch angle by OUC glider

The ALBAC glider used its payload to glide downward and upon releasing its payload, glided upward [2]. However most gliders such as Slocum [22], Spray [23] and Seaglider [18] use a buoyancy engine instead. These three gliders are fixed winged buoyancy propelled gliders which shift their internal ballast mass to control their attitude [1].

Leonard and Graver [1] developed a laboratory scale ROGUE glider to implement a model based feedback open loop approach. Another laboratory scale glider, the ALEX glider, developed by Arima et al. [7] designed an independently controllable wings and rudder (Fig. 2).

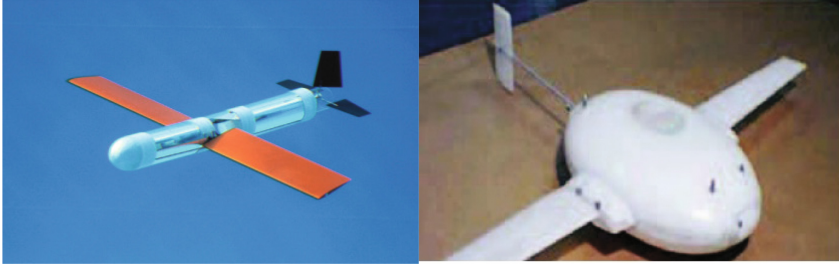


Fig. 2. Laboratory scale gliders, ALEX glider [7] and ROGUE glider [1]

3 Control Strategies

Several control strategies have been proposed to control the position, velocity and trajectory of UGs such as PID controllers by Bhatta and Noh [24, 25], Linear Quadratic Regulator (LQR) by Nina and Graver [9, 26], Sliding Mode Controllers (SMC) by Yang and Song [27, 28], Neural Network (NN) by Isa [19] and Dong [29], Fuzzy logic for self-tuning by Loc [30] and Model Predictive Controllers by Cooney [31] and Steenson [32]. PID control loops used for pitch and roll control by Panish [16] allows their UG to be guided in a variety of ways. In such control strategies, vertical and horizontal modes are decoupled to control the pitch, angle of attack and depth of the glider. The robustness of PID controller using nonlinear time invariant aspects of glider designs is investigated by Bender [33]. AUG dynamics are nonlinear but PID controllers are based-on linear control law. Therefore, PID controller cannot dynamically compensate for un-modeled vehicle hydrodynamic forces and unknown disturbances. As such, combinations of control strategies to form hybrid controllers are used to handle the complexity that is produced due to nonlinearity and hydrodynamics effect.

Fuzzy-PID controller for underwater glider pitch control has been used by Liu and Su [51]. Fuzzy-PID can adjust the control parameters to adapt to the change of environment. It can keep the stability of underwater glider. However, fuzzy rules, fuzzification and defuzzification are very complex.

Sliding mode control (SMC) for underwater trajectory tracking control has been used by Yang and Bessa [27, 34]; and Liu [21]. SMC is a robust system with low sensitivity to environmental disturbances. However, chattering is a well known downside of this control strategy, which can degrade the performance of gliders and may lead to instability.

A multivariable Multiple input Multiple Output (MIMO) control system have been proposed for AUVs and ROVs including self-tuning and position control by Goheen [35]. The dynamic model using LQR technique was implemented in a laboratory scale ROGUE glider in order to examine its motion. However, LQR has poor switching performance between upward and downward glide paths [26].

In a comparative study, Nag [36] discussed the use of an adaptive fuzzy control strategy for tracking trajectory system combined with PID controller. However, they considered the glider as a SISO system only and neglected nonlinearities. A recurrent NN based on MPC control strategy for motion control problem in longitudinal plane is proposed by Shan [37] for convergence to optimal solution. However, in this approach the effect of water current has not been considered.

Due to the inherent non-linearity in the dynamics, and hence difficulties in determining an accurate mathematical model of the vehicle, an adaptive fuzzy logic based control strategy coupled with dynamic compensators have been designed. The effect of fuzzy logic for depth controller has been implemented on KAUV-1 which has the advantage of fast self-tuning [30]. The simulation results showed satisfactory performance but again, the effect of external disturbance and nonlinearity were not investigated. The nonlinearity of the Odyssey IV for the purpose of disturbance rejection using Model predictive controller was discussed by Cooney [31] and a hybrid PIDMPC controller is suggested for improving and predicting future values. However, a communication problem between commanded input and actual input was observed during implementation of this strategy in Odyssey IV. In [27], Yang addressed the robustness issue produced due to nonlinearities and proposed the adaptive sliding mode controller. However, the resulting chattering effect lead to instability.

Neural network (NN) control strategy [19] have been proposed as a solution to the lack of robustness due to nonlinear effect in dynamic models. Due to the ability to handle nonlinearity and adapt to changing conditions as is the case of varying currents in glider dynamics, NN has been implemented in USM hybrid underwater glider [19]. In proposed model, all glider parameters e.g. pitch, speed, maneuverability, sliding mass, wing, rudder and propeller were controlled. A neural network self-tuning PID controller [29] for a spherical AUV was proposed for the purpose of controlling the vehicle velocity and tracking the desired target in the vehicle fixed coordinate system.

4 Hybrid Control Strategy

The performance of individual control strategies can be improved by combining it with another control strategy to form a hybrid control strategy. Different implementations of this concept have been applied successfully in AUVs such as Adaptive fuzzy sliding mode control strategy.

4.1 Combining Fuzzy Logic and Sliding Mode

Fuzzy logic has been used to design a sliding mode controller [28, 38]. However, the FLC was implemented in a SISO system only. Alternatively, PID and FLC have been

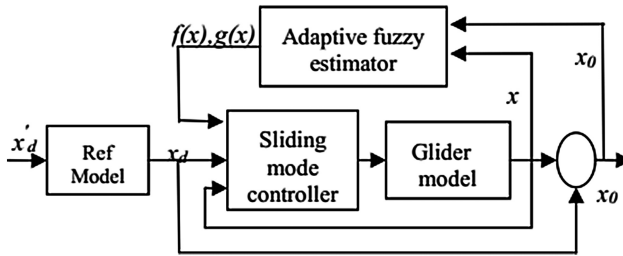


Fig. 3. Adaptive fuzzy sliding mode control [38]

combined to design a new controller. To eliminate the chattering effect of sliding mode, a boundary layer is introduced around the sliding surface by Song [28]. A general block diagram of adaptive fuzzy sliding mode controller is shown in Fig. 3.

4.2 Sliding Mode Fuzzy Controller

In this approach, sliding mode is used to design functions of the fuzzy logic controller and a switching function is formed which can eliminate the chattering effect. This type of approach was experimentally applied to OEX series AUVs for pitching and heading control. The inputs to SMFC heading controller were defined as heading error while the output for heading controller was to control the rudder [39].

The control strategy can be further improved using a fuzzy sliding mode controller has only been investigated for combination of nonlinear controllers for the motion of the Snorkel AUV [38]. Recent work concerning different autonomous underwater gliders based on added mass and control techniques for AUVs and AUGs is summarized in inertia, effect of external varying disturbances and rudder Table 2 control inputs [40]. However the effectiveness of an adaptive.

5 Major Limitation and Challenges

Very few control schemes are proposed to measure the sensitivity analysis of environmental disturbances such as Isa [12] and Fan [50], but this is a new idea in undersea vehicles to consider both water current estimation and payload factor in dynamic modelling and control for UGs. The main attempt is to choose a controller that can guarantee the suppression or at least the limitation of the overshoot in the system response. An appropriate control strategy is required that will damp out the external disturbances and will follow trajectory with minimum error for varying payloads.

Table 2. Control strategies

Controller design	Implementation	Advantages	Limitation
PID control [41, 44]	Bluefin AUV [16] Seaglider [45], SLOCUM glider ALEX glider	The system adopts itself in a limited time and has the ability to decouple the vertical and horizontal modes. [16] PID controller regulator is used to follow the desired trajectory for Seaglider	The AUG dynamics are nonlinear and that a PID controller uses a linear control law and cannot provide easy control implementation due to nonlinear dynamics: It is difficult to tune the system with a single set of control gains to respond equally in both directions [16]
LQR [46] feed forward and feedback control	ROGUE Laboratory scale glider [47],	Using this method, comparatively less tuning is required. It has the ability to handle the MIMO system through state space representation	This is applicable to linear control laws only. This method is difficult to implement on unsteady motion controls
Neural network control [19, 29]	USM hybrid underwater glider, PETREL, hybrid underwater glider	It has the ability to handle the MIMO system with low rates of control update. Two planes, vertical and horizontal, can be controlled using a single controller [19]. Fast convergence and self-tuning effect	Cannot meet the requirement of rapid response. Complex for real time application
Sliding mode control [27, 39]	NPS ARIES AUV Grace(Gliding robotic fish)	Stability of SMC is in-built. Invariance property and ability to decouple high dimensional system into subsystems with lower dimensionality	This can easily lead to system chattering and effect control accuracy. Limited disturbance effect was included in the model [27]
Fuzzy logic control [36, 39]	KAUV-1 [36]	Needs expert knowledge to optimize. Easily implemented on several AUVs,	The stability and overshoot problem occurred at multiple points, including tuning fuzzy rules [48].
Combination of fuzzy logic and sliding mode AFSMC [28]	OEX Series AUVs Snorkel AUV [38]	The errors found in FLC and SMC is easily eliminated by using combined effect of both of these. Chattering effect is minimized [34]	This was implemented on experimental AUV for limited depth. [28]

(continued)

Table 2. (continued)

Controller design	Implementation	Advantages	Limitation
Nonlinear model predictive control (MPC)	Odyssey IV (AUV) USM	MPC vs PID controller comparison and PID-MPC hybrid controller is designed for minimum cost. It has the ability to handle constraints and nonlinear systems	There was a disconnect problem between actual input and command input due to communication problem with the software during implementation of MPC
Fuzzy-PID control [51]	OUC glider	Fuzzy-PID controller can solve non-linear and linear control problem. It has the ability to finish the control of complex system	Fuzzy control rules, fuzzification and defuzzification are very complex
Nonlinear, passivity-based controller [13, 49]	Miniature UWG Grace(Gliding robotic fish)	Fast convergence, Easy implementation	

6 Conclusion

Different control strategies have been demonstrated for acceptable control performance in different AUVs and AUGs, but they still have limitations in compensating for the high nonlinearity of underwater gliders including external disturbances and carrying large payloads which are the main reasons that make it a challenging problem to control the underwater glider. Due to this it is highly desirable to design a controller that is robust and will give real time response. For this, combinations of two or more controllers have been developed by some researchers for AUVs to form a hybrid controller which can minimize the negative effect and will improve the performance.

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